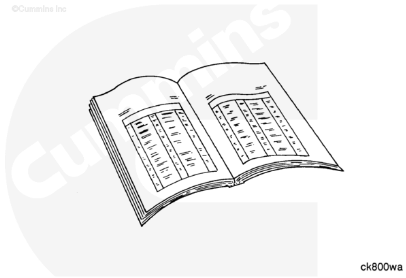
Trainings

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

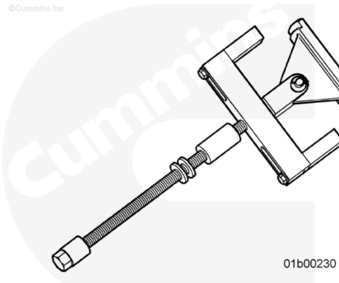
- Disconnect the battery. Refer to the equipment manufacturer service information.
- Drain the coolant. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)
- Drain the oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/20/20-007-037.html)

- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/20/20-002-004-tr.html)
- Remove the piston and connecting rod. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-054-tr.html)

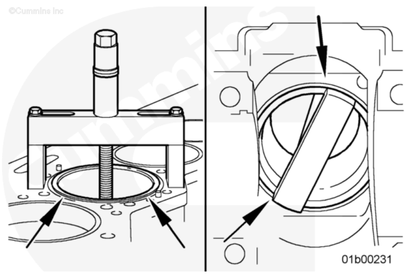
Remove

Use the cylinder liner puller, Part Number 3163745, and puller plate, Part Number 3162886, to remove the cylinder liner.

Wind down the threaded rod to extend the arm. Turn the arm to the side to enable passage through the liner.

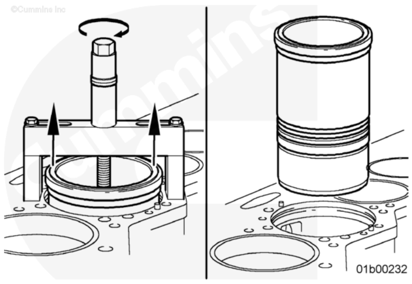


Install the cylinder liner puller, Part Number 3163745, in the cylinder liner. The puller feet **must not** touch the top of the liner. The puller arms **must** be positioned firmly on the bottom of the liner.



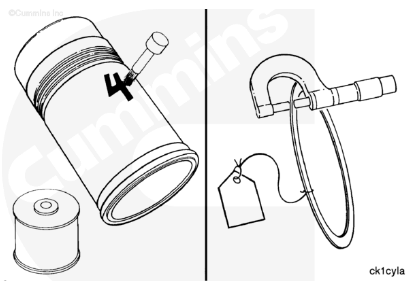
Turn the puller screw until the liner loosens in the block.

Remove the tool and the liner.



Use a liquid metal marker to mark the cylinder number and bank on each liner. Mark the cylinder liner on the camshaft side of the cylinder liner.

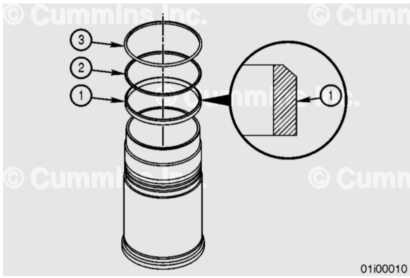
If sealing rings were used, use a tag to mark the cylinder number.



Measure in several places and record the thickness of the sealing rings used in each cylinder. The thickness of the sealing ring is one factor in determining liner protrusion. This information **must** be known when the liners are installed in the engine.

Remove and discard the two D-rings (2 and 3).

Remove and discard the crevice seal (1).



Clean and Inspect for Reuse

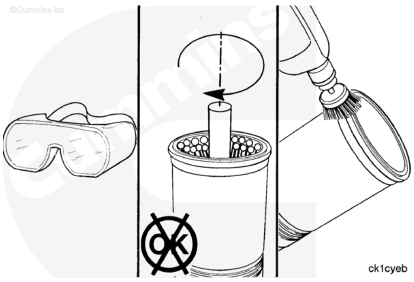
⚠ WARNING ⚠

Wear appropriate eye and face protection. Flying dirt and debris can cause personal injury.

⚠ CAUTION ⚠

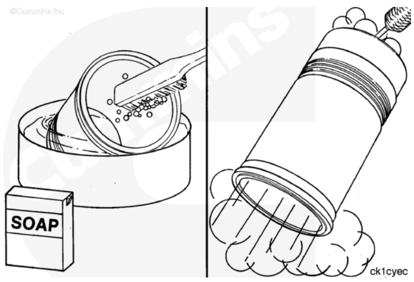
Do not use a hone, deglazing, or prebrushing to clean the cylinder liners. Abrasives can damage the finish and the crosshatch pattern and can contaminate the liner.

Use a high quality steel wire brush to clean the liner flange seating area. Make sure the wire brush is rated for the rpm being used if the brush is motor driven.



⚠ WARNING ⚠

When using a steam cleaner, wear protective clothing and safety glasses or a face shield. Hot steam can cause serious injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not contaminate the wash tank solution with bead blast materials. Use a kerosene emulsion-based solvent. Do not use a solvent that contains chlorinated hydrocarbons with cresol, phenols, or cresylic components.

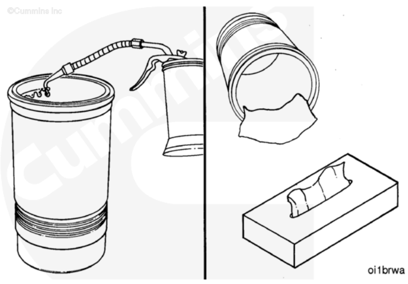
Use a nonmetallic bristle brush, detergent soap, and warm water to clean the inside of the liner.

Clean the liners with solvent or steam.

Dry with compressed air.

Use clean engine oil to lubricate the inside diameter.

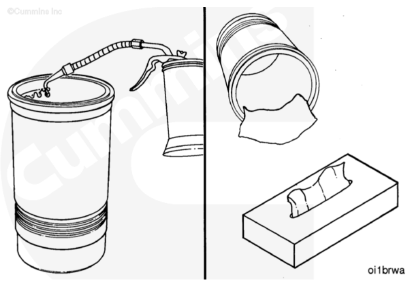
Allow the oil to soak the liner inside diameter for 5 to 10 minutes.



oi1brwa

⚠ CAUTION ⚠

Do not use a cloth towel. Lint will cause severe engine damage.



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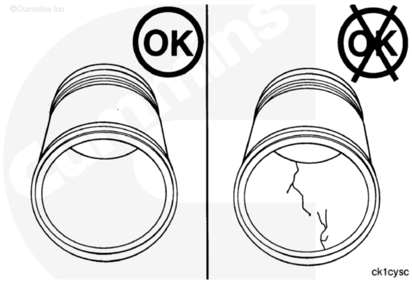
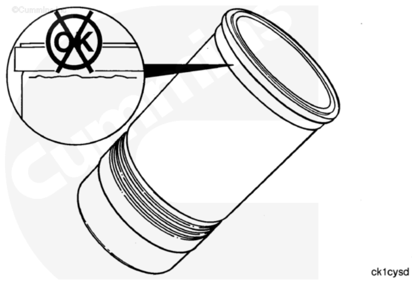
Wipe the liner inside diameter with a paper towel.

Repeat these three cleaning steps until the paper towel shows **no** gray or black residue.

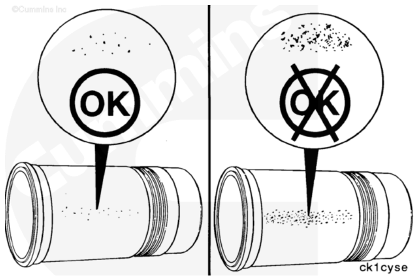
Inspect the liners for cracks on the inside diameter and the outside surfaces.





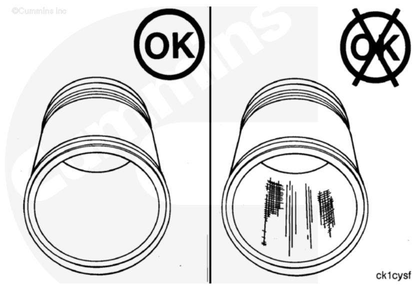
Note : Cracks can also be detected by using either magnetic inspection or the dye method.

Inspect for cracks under the flange.



Inspect the outside surface for excessive corrosion or pitting. Pits **must not** be more than 1.6 mm [0.063 in] deep.

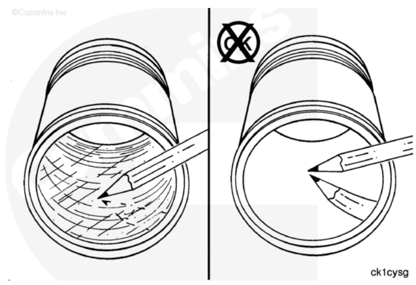
Replace the liner if the pits are too deep or if the corrosion can **not** be removed with a fine emery cloth.



Note : If a fingernail catches in the scratch, the liner must be replaced.

Inspect the inside diameters for vertical scratches deep enough to be felt with a fingernail.

Inspect the inside diameter for scuffing or scoring.



Inspect the inside diameter for liner bore polishing.

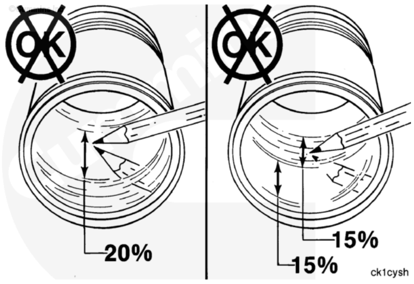
A moderate polish produces a bright mirror finish in the worn area with traces of the original hone marks or an indication of an etch pattern.

A heavy polish produces a bright mirror finish in the worn area with no traces of hone marks or an etch pattern.

Reference the Parts Reuse Guidelines, Bulletin 3810303, for further information on liner bore polishing.

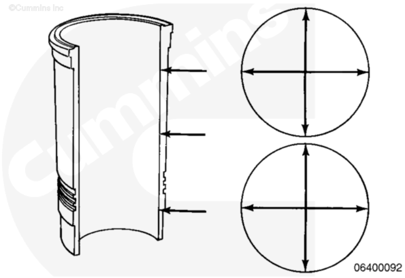
Replace the liner if:

- A heavy polish is present over 20 percent of the piston ring travel area.
- Both moderate and heavy polish is present over 30 percent of the piston ring travel area, with one-half of the 30 percent (15 percent) consisting of heavy polish.



Note : The inside diameter of a new cylinder liner can be 0.015 mm [0.0006 inch] smaller than specifications because of the Lubrite™ coating.

Use a dial bore gauge to measure the inside diameter of the liner at the top, the bottom, and the middle of the piston ring travel area. Perform two measurements at each location. The measurements **must** be 90 degrees apart.

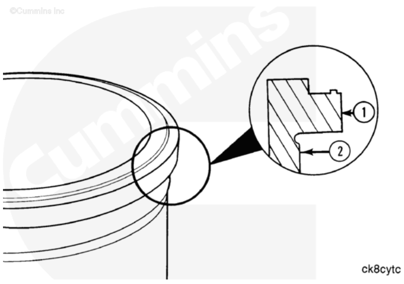


Cylinder Liner Inside Diameter		
mm		in
158.737	MIN	6.2495
158.877	MAX	6.2550

The cylinder liners are:

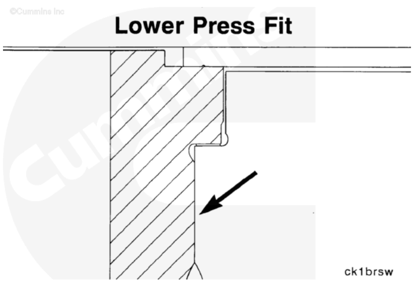
1. Flange outside diameter referred to as the upper press fit outside diameter.
2. Liner outside diameter referred to as lower press fit outside diameter.

For example, 20/20 OS means the upper press fit outside diameter (1) is 0.020 inch larger than the standard liner and the lower press fit outside diameter is 0.020 inch larger than the standard liner.

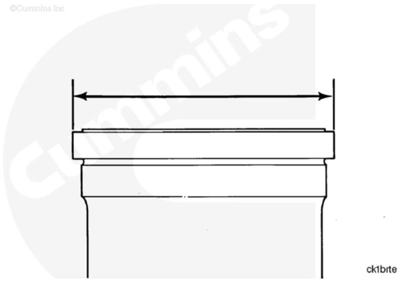


The liner design incorporates a press fit between the upper liner bore and the area of the liner directly below the liner flange. This is referred to as the lower press fit design.

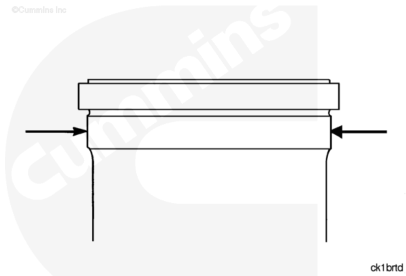
Cylinder liners with standard and oversize press fit diameters are available.



Measure the liner flange outside diameter.



Measure the lower press fit diameter. Reference the following tables for liner sizes.

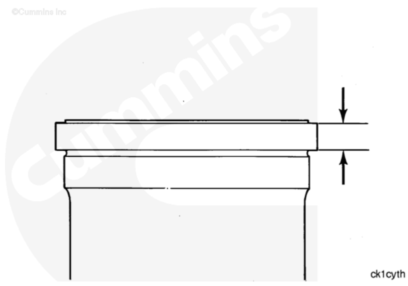


Mechanically Actuated Injectors							
		Upper Press Fit Liner Flange Diameter			Lower Press Fit area Outside Diameter		
Part Number	Size (in)	mm		in	mm		in
4333984	Undersize -0.01/0	187.965	MIN	7.400197	180.165	MIN	7.09311
		188.189	MAX	7.402165	180.215	MAX	7.095079
4308809	Standard	188.697	MIN	7.409016	180.165	MIN	7.09311
		188.239	MAX	7.410984	180.215	MAX	7.095079
4371641	Oversize +.010/.010	188.443	MIN	7.409016	180.419	MIN	7.09311
		188.493	MAX	7.420984	180.469	MAX	7.105079
4333987	Oversize +.020/.020	188.697	MIN	7.429016	180.671	MIN	7.113031
		188.747	MAX	7.430984	180.721	MAX	7.115
4371767	Oversize +.030/.030	188.951	MIN	7.439016	180.927	MIN	7.125079
		189.001	MAX	7.44-984	180.77	MAX	7.125079

Mechanically Actuated Injectors							
		Upper Press Fit Liner Flange Diameter			Lower Press Fit area Outside Diameter		
Part Number	**Size (in)**	**mm**		**in**	**mm**		**in**
4371642	Oversize +.040/.040	189.205	MIN	4.449016	181.181	MIN	7.13311
		189.255	MAX	7.450984	181.231	MAX	7.135079
4333989	Oversize +.060/0	189.255	MIN	7.450984	181.231	MIN	7.135079
		189.713	MAX	7.469016	180.165	MAX	7.09311
4333988	Oversize +.060/.020	189.713	MIN	7.470984	180.671	MIN	7.113031
		189.763	MAX	7.470984	180.271	MAX	7.115

Measure the liner flange thickness.

Electronically Actuated Injectors Cylinder Liner Thickness			
	mm		in
Undersized Liner	7.747	MIN	0.305
	7.773	MAX	0.306024
Standard and Oversized Liner	13.398	MIN	0.52748
	13.424	MAX	0.528504

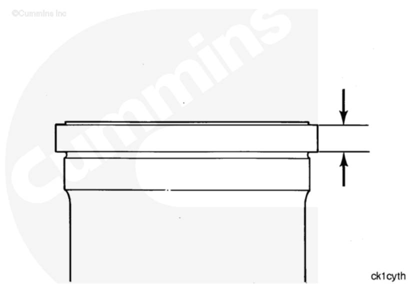


Electronically Actuated Injectors							
		Upper Press Fit Liner Flange Diameter			Lower Press Fit area Outside Diameter		
Part Number	**Oversize (in)**	**mm**		**in**	**mm**		**in**
4333993	Standard	188.189	MIN	7.409016	180.165	MIN	7.093110
		188.239	MAX	7.410984	180.215	MAX	7.095079

Electronically Actuated Injectors							
		Upper Press Fit Liner Flange Diameter			**Lower Press Fit area Outside Diameter**		
Part Number	**Oversize (in)**	**mm**		**in**	**mm**		**in**
4333993	Oversize +.010/.010	188.443	MIN	7.419016	180.419	MIN	7.103110
		188.493	MAX	7.420984	180.469	MAX	7.105079
4333996	Oversize +.020/.020	188.697	MIN	7.429016	180.671	MIN	7.113031
		188.447	MAX	7.430984	180.721	MAX	7.115000
4333994	Oversize +.030/.030	188.951	MIN	7.439016	180.927	MIN	7.123110
		189.001	MAX	7.440984	180.977	MAX	7.125079
4333995	Oversize +.040/.040	189.205	MIN	7.449016	181.181	MIN	7.133110
		189.225	MAX	7.450984	181.206	MAX	7.134094

Measure the liner flange thickness.

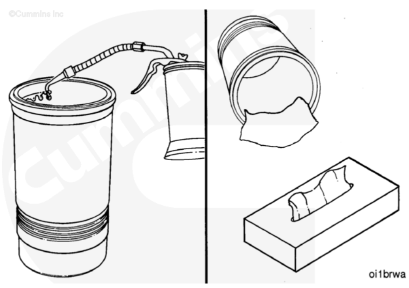
Electronically Actuated Injector Cylinder Liner Flange Thickness		
mm		in
13.398	MIN	0.52748
13.541	MAX	0.53311



Do **not** place the liners in an area where dirty air flow can contaminate the liners.

Apply a thick film of clean 15W-40 oil to the bores of the liners for final cleaning. Leave the oil on for 5 to 10 minutes.

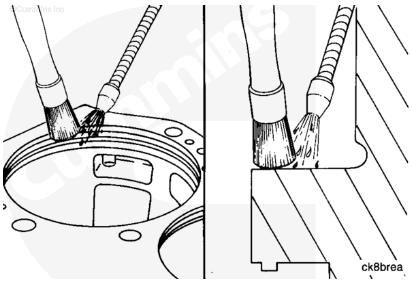
Use a clean, lint-free paper towel to wipe the oil from the bores until the black and gray deposits are removed.



Install

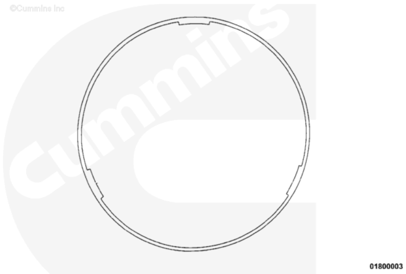
⚠ WARNING ⚠

When using solvents, acids, or alkaline material for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



Clean the bottom of the cylinder liner flange with safety solvent.

Note : The seal rings have three locating tabs on the inside diameter. The tabs have an interference fit to the liner lower press fit diameter, to hold the seal ring in place during liner installation.

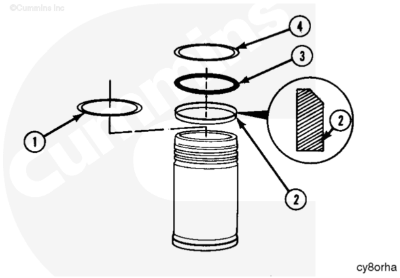


Install the seal rings.

The seal ring **must** be straight on the liner upon installation. Use your fingers to push the seal ring near the tabs to fit the seal ring down and over the lower press fit diameter during installation.

This practice during installation of the seal ring will prevent deformation that will result in the seal ring **not** fitting squarely on the bottom of the liner flange.

Note : Some o-rings have a "D" shape cross section. This type of o-ring must be installed with the flat side against the cylinder liner.



Install the crevice seal. The beveled edge of the crevice seal (2) **must** be positioned as shown.

Install o-rings in the position shown. Use the mold mark on the o-ring to check if the o-ring is twisted.

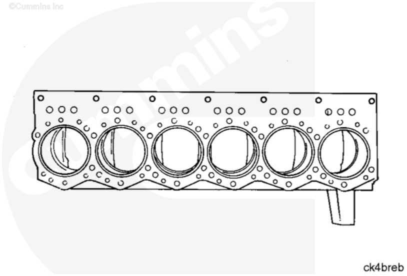
1. Liner, counterbore sealing ring
2. Crevice seal
3. Black o-ring
4. Red o-ring.

⚠ CAUTION ⚠

Do not use petroleum-based oil. This will cause swelling of the packing ring.

Use vegetable oil to lubricate the inside diameter of the cylinder block packing ring bores.

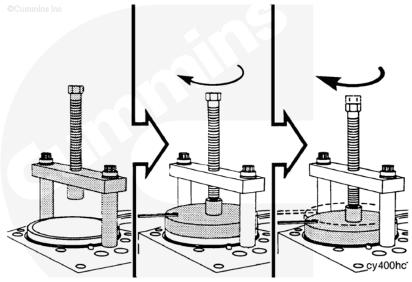
Push the cylinder liners into the block with hand pressure.



Use liner installation tool, Part Number, 3375422, or equivalent to install the bridge assembly and two cylinder head capscrews. Tighten the capscrews.

Torque Value: 65 n•m [48 ft-lb]

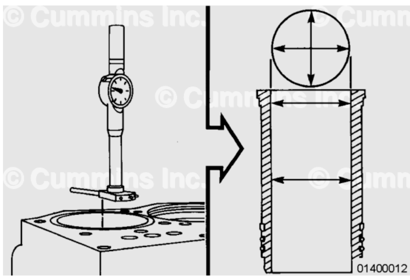
Install the pusher plate in the liner. Make sure it is aligned correctly in the liner. Turn the pusher screw until it touches the plate. Turn the pusher screw until the liner flange touches the counterbore ledge. Do **not** use more than 65 N•m [48 ft-lb] of torque. Remove the tool.



Note : New cylinder liners can be 0.005 mm to 0.015 mm [0.0002 in to 0.0006 in] smaller than the minimum specifications because of the Lubrite™ coating.

Use a dial bore gauge and measure the inside diameter of the liner at the top, bottom, and middle of the liner.

Perform two measurements at each location. The measurements **must** be 90 degrees apart.



New Cylinder Liner Inside Diameter		
mm		in
158.737	MIN	6.2495
158.775	MAX	6.2510

The inside diameter **must** be no more than 0.076 mm [0.003 in] out-of-round at the top two measurements.

If the inside diameter is more than 0.05 mm [0.002 in] out-of-round in the bottom measurement location, the liner **must** be removed. Check for a twisted o-ring.

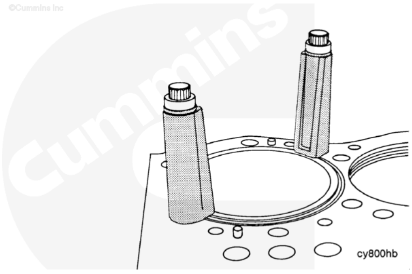
Use the following procedure if machining is required. Refer to Procedure 001-058 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-058.html)

Measure

The clamps **must** touch the highest part of the liner.

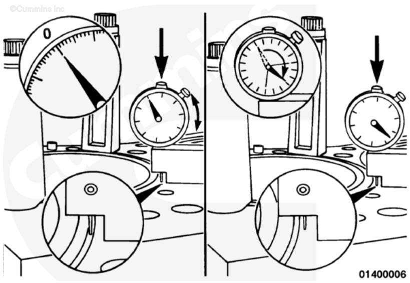
Use two cylinder liner clamps, Part Number 3376944, and two cylinder head capscrews. Install the clamps and capscrews. Tighten the capscrews.

Torque Value: 65 n•m [48 ft-lb]



Use a depth gauge block, Part Number 3823495, or equivalent. Position the depth gauge block so the indicator needle will contact the liner flange on the outside of the sealing bead, as shown.

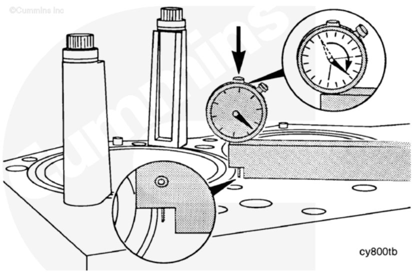
Gently push the indicator needle down until it touches the liner. Turn the gauge until the "0" is aligned with the head of the dial. Repeat this step several times to be sure the gauge is set to "0".



Raise the indicator and move the gauge block until the indicator will touch the block surface. Gently push the indicator down until it touches the block. Read the indicator.

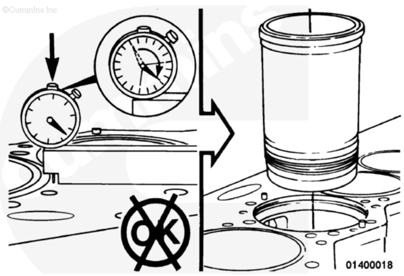
Cylinder Liner Protrusion		
mm		in
0.13	MIN	0.005
0.18	MAX	0.007

Read the liner protrusion in four places equally spaced on the liner outside diameter. The measurements on each liner **must not** vary more than 0.03 mm [0.001 inch].



If the liner protrusion is **not** correct, the liner **must** be removed.

Use seal rings to increase the protrusion. Machine the counterbore ledge to decrease the protrusion. Use the following procedure if machining is required. Refer to Procedure 001-058 in Section 1 (/qs3/pubsys2/xml/en/procedures/20/20-001-058.html)



Finishing Steps

⚠ **WARNING** ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ **WARNING** ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ **WARNING** ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ **WARNING** ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/20/20-001-054-tr.html)
- Install the cylinder head and related components. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/20/20-002-004-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html)

- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/20/20-007-037.html)
- Connect the battery. Refer to the equipment manufacturer service information.
- Operate the engine and check for leaks.
- Perform an engine run-in, if necessary. Refer to Procedure 014-005 in Section 14. (/qs3/pubsys2/xml/en/procedures/20/20-014-005.html)

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